



Changing a data point

1 Which of these statements defines the median? Tick **1** correct answer

- ☐ The most frequent value when the data are put into numerical order.
- ☐ The distance on a number line between the smallest and largest data point.
- ☐ The sum of values divided by the number of values in a data set.
- ☒ The middle piece of data when the data are put into numerical order.

2 Calculate the mean for this data set. Tick **1** correct answer

4	7	16	16	22	23	24	28
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- ☐ 7
- ☐ 16
- ☒ 17.5
- ☐ 19
- ☐ 24



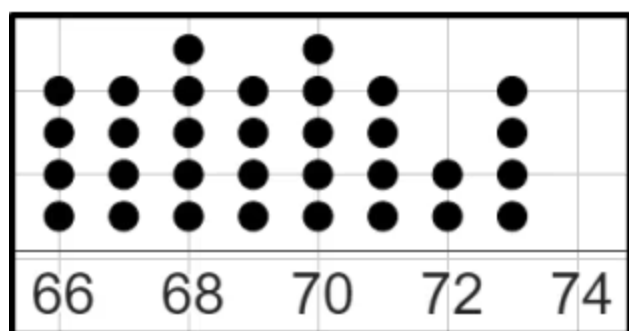
3 Match the summary statistic to its value. Write the correct letter in each box

-2	-1	-1	4	5	6	10
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a	mean
b	median
c	mode
d	range

<i>b</i>	4
<i>d</i>	12
<i>a</i>	3
<i>c</i>	-1

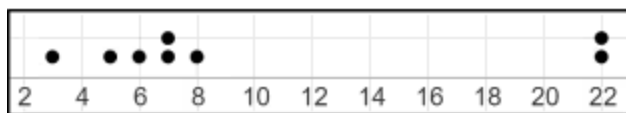
4 Which of these statements is true for this dot plot? Tick 3 correct answers



- ☐ The data set has a single mode.
- ☒ The data set is bimodal.
- ☒ The mode does not appear to be the most representative summary.
- ☐ Because the mode is not representative, it is not a useful summary.
- ☒ Even though the mode isn't representative, this information can still be useful.

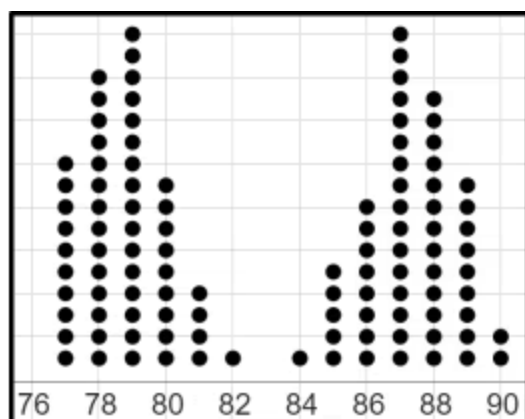


- 5 First calculate the mean for this dot plot, then select the correct statements. Tick **2** correct answers



- ☒ The mean is 10.
- ☐ The mean is 19.
- ☐ The mean is representative because many data points share the same value.
- ☒ The mean isn't very representative because not many data point are close to it.
- ☐ The mean isn't very representative because the range is bigger.

- 6 Without carrying out any calculations select which of these statements are likely to be true for this dot plot. Tick **2** correct answers



- ☐ The median is 13.
- ☐ The median is 78.
- ☒ The median is 83.
- ☐ The median is close to the two peaks of high frequency values in the data set.
- ☒ The median is in the dip of low or no frequency values in the data set.

